3 (a) (i) Taking direction to the right of the equilibrium position as positive

By Newton's 2nd Law,

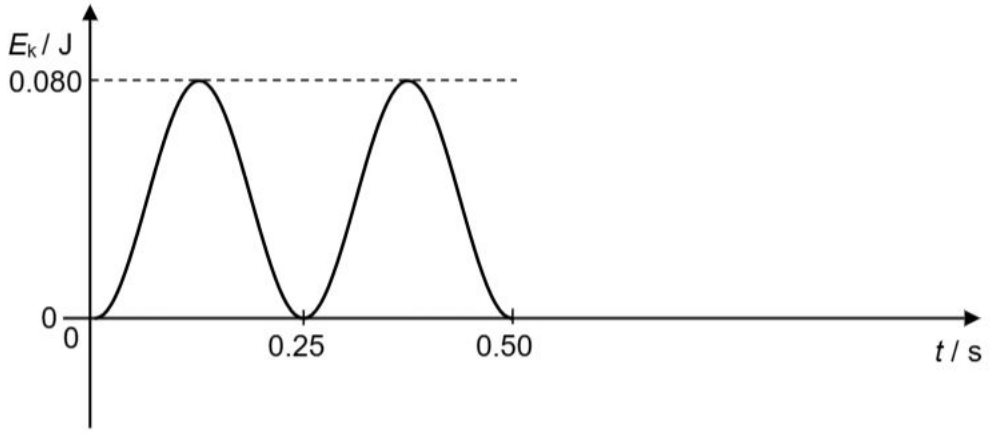
$$F_{net} = ma$$
$$k(e-x) - k(x+e) = ma$$
$$-2kx = ma$$
$$a = -\frac{2k}{m}x$$

(ii) 1. For an object moving in SHM, $a = -\omega^2 x$

Since $\omega^2 = \frac{2k}{m}$,

$$T = \frac{2\pi}{\omega} = \frac{2\pi}{\sqrt{\frac{2k}{m}}}$$
$$= \frac{2\pi}{\sqrt{\frac{2(50)}{0.63}}}$$
$$= 0.4987 = 0.50 \text{ s}$$

2.
$$v_o = \omega x_o = \sqrt{\frac{2k}{m}}x_o$$
$$= \sqrt{\frac{2(50)}{(0.63)}}(0.040) = 0.50395 = 0.50 \text{ m s}^{-1}$$



$$E_{k,\max} = \frac{1}{2}mv_o^2 = \frac{1}{2}(0.63)(0.50395^2) = 0.079999 = 0.080 \text{ J}$$

*correct shape with smooth turning points, correct $E_{k,\max}$ and period.

(b) For the same applied force, the new composite spring will stretch by double the original amount. Hence, the effective force constant will be half that of the single original spring.

Consequently, the angular frequency will decrease by a factor of $\sqrt{2}$ and since the period is inversely proportional to angular frequency, the period will increase by a factor of $\sqrt{2}$.

OR

At the same displacment, the restoring force is halved as the effective force constant will be half that of the single original spring.

Consequently, the angular frequency will decrease by a factor of $\sqrt{2}$ and since the period is inversely proportional to angular frequency, the period will increase by a factor of $\sqrt{2}$.